



Statement of Work ARC Application Implementation

This Statement of Work ("**SOW**") is incorporated by reference into and subject to the terms and conditions of the **Enterprise Sourced Talent Agreement**, as may be amended from time to time, between Toptal, LLC ("**Toptal**") and Syngenta Crop Protection, LLC ("**Client**") with an effective date on or about November 22, 2019 ("**Agreement**"). This SOW becomes effective on October 7, 2026 ("**SOW Effective Date**"). The Agreement will take precedence and will govern over any inconsistent or conflicting term(s) in this SOW, unless and solely to the extent that the parties expressly state below in this SOW that they intend to override a specific term of the Agreement. For the avoidance of doubt, this SOW is considered a "TOP" as defined under the Agreement. Capitalized terms used herein and not otherwise defined will have the meanings given them in the Agreement.

1. Scope of Work.

- A. Toptal will perform Work and provide Work Output in accordance with the specifications set forth in Exhibit A (Arc Genomics Platform - Implementation Specifications), attached hereto and incorporated by reference into this SOW. In the event of any conflict between this SOW and Exhibit A, this SOW will control unless expressly stated otherwise.
- B. The Toptal Delivery Manager is Saadia Iqbal. Toptal will notify Client promptly of any change to the Delivery Manager during the SOW term.

2. Fees.

- A. Toptal will perform the Scope of Work for Client on a **Fixed Fee (milestone billing)** basis, invoiced upon completion of the milestone. If the actual completion date differs from the Estimated Completion Date specified below, Client will be notified in writing (email to suffice).
- B. Based on the Scope of Work, timing, Client responsibilities, and the assumptions and dependencies set forth herein, the **milestone fee schedule** under this SOW will be as set forth in the table below.

Milestone	Fee	Estimated Completion Date
Milestone 1: P1 Foundation	\$26,808	November 17, 2026
Milestone 2: P2a Services and Data	\$60,356	January 26, 2027
Milestone 3: P2b Application Surface	\$68,210	March 2, 2027
Milestone 4: P3 Launch	\$44,991	April 6, 2027

- C. Based on the Scope of Work, timing, Client responsibilities, and the assumptions and dependencies set forth herein, the **total Fixed Fee** under this SOW will be **\$200,365**.

3. Schedule.

- A. Estimated Start Date October 7, 2026.
- B. Estimated Completion Date April 6, 2027.
- C. Adjustment. If, following execution of this SOW, the Estimated Start Date is changed by mutual agreement of the parties (whether advanced or delayed), the Estimated Completion Date will shift correspondingly so that the estimated period of performance contemplated by this SOW remains unchanged. Such an adjustment does not require a Change Order; the parties will confirm the revised dates by email or other written agreement.

4. Assumptions and Dependencies.

Any and all project goals, timelines, requirements, estimated hours and fees are dependent upon the assumptions and dependencies set forth below (collectively, “**Assumptions**”):

- Estimates are based on initial discussions with the Client and information provided as of the SOW Effective Date.
- Key Client stakeholders will be available to attend meetings as outlined in Exhibit A.
- The Client Contact will provide timely feedback, reviews, approvals, and answers to questions, and will provide Toptal with access to all necessary documents, codebases, services, tools, and associated elements as quickly as possible or as outlined in Exhibit A.
- The Client will not exercise a buyout option of any Talent providing some or all of the Scope of Work hereunder during the term of this SOW.
- Compliance Responsibility: Client is solely responsible for ensuring the legal and regulatory compliance of the product, including, without limitation, defining any and all project requirements to comply with applicable law or standards. Client agrees to indemnify, defend, and hold Toptal harmless against any claims related to the project’s compliance with applicable laws, regulations, or other standards.
- Training and related training documentation are out of scope for this engagement unless otherwise specified in Exhibit A.
- Changes that may significantly impact scope, timeline, or project costs will be addressed promptly with the Client and may require a Change Order.
- The assumptions and dependencies and the Client responsibilities set forth in Exhibit A are incorporated into, and form part of, the Assumptions under this Section 4.

The project goals, timelines, requirements, estimated hours and/or fees will be equitably adjusted by the parties in good faith to address any inaccuracies or insufficiencies in any of the Assumptions. Each party will provide the other party with written notice (email or other electronic means sufficient) if it discovers that any portion of the Assumptions are inaccurate or incomplete and the parties will promptly cooperate to make the needed adjustments by executing an amendment or change order to this SOW or by such other written means (*e.g.*, email) as may be agreed by the parties.

5. Third Party Items.

The third party tools, applications, utilities, platforms, and/or cloud infrastructure required for the performance of the Scope of Work and/or used to produce Work Output (collectively, "***Third Party Items***") will be provided or made available by the party indicated in Exhibit A. If a Third Party Item can only be provided or made available pursuant to a new agreement or an amendment to an existing agreement, then Toptal and Client will use commercially reasonable good faith efforts to negotiate and execute the same. Third Party Items provided or recommended by Toptal are "As Is" and without any warranty or indemnity of any kind from Toptal and Toptal will not have any liability for any delay or failure of all or a portion of the Third Party Items.

Where Work is performed or Work Output is produced in a Third Party Item (e.g., a third-party design, collaboration, or development platform), Client is responsible for procuring and maintaining any licenses required to access, view, or edit such Work or Work Output following delivery. Toptal will provide access to any such Work Output during the engagement at no additional cost to Client.

6. Client Contact. ***Mike Magwire***

The Client Contact is responsible for allocating reasonable time, attention, effort and timely cooperation to reasonably support Toptal's performance of the scope of work under this SOW. The Client Contact will participate in planning, working, and review sessions throughout the engagement, will be available for interim and final reviews, and will be responsible for Acceptance and Sign-Off of this SOW as set forth in Section 7. If material concerns are raised during the Acceptance Period, the Client Contact will participate in the working session contemplated under Section 7 to resolve such concerns in good faith.

If the Client Contact becomes unavailable for any extended period of time, Client will promptly notify Toptal and will use commercially reasonable efforts to promptly provide a knowledgeable and suitable replacement. Client acknowledges and agrees that any failure by Client to fulfill its obligations under this Section will have a direct impact on Toptal's ability to timely complete the affected scope of work and will cooperate with Toptal to equitably adjust this SOW (including schedules and fees) to account for any such failure by Client to fulfill its obligations under this Section.

7. Acceptance.

The evaluation and acceptance process set forth in this Section will apply to **each Milestone defined in Section 2 "Fees" above**. Client will have five (5) business days from the date of notice of Toptal's completion of the Milestone, as applicable (the "***Milestone Acceptance Period***"), to evaluate, review and/or test the Work Output (or portion thereof) to determine if it meets the corresponding specifications in all materials respects. *An Acceptance Criteria (including one expressed as 100%) is deemed satisfied if the Work Output materially conforms to the criterion and any deviation does not materially affect its functionality, performance, security, or usability.* If Client does not provide written notice to Toptal specifying that a specific Milestone (or portion thereof) has failed to satisfy its specifications in all material respects prior to the end of the Acceptance Period and/or if Client commences using the Work Output commercially or in a production environment, then Client will be deemed to have accepted the Milestone (or the portion of the Work Output that corresponds to the applicable milestone). Client will sign and deliver to Toptal a certification of milestone acceptance. If any Milestone (or portion thereof) fails to satisfy its

corresponding specifications in any material respect, then Client will notify Toptal in writing specifying the ways in which the Work Output (or portion thereof) does not conform and what modifications are required to bring it into conformance. Toptal will then use its commercially reasonable efforts to modify the Work Output (or portion thereof) under the corresponding Milestone to bring it into conformance and will resubmit the Work Output (or portion thereof) to Client for acceptance. If after three (3) repeated attempts, Toptal is unable to remedy within Client's reasonable discretion, in all material respects, any non-conforming portion of any Work Output, Client may terminate this SOW pursuant to the terms of the Agreement. On any such termination, Client will pay Toptal all amounts due or accrued under this SOW through to the termination date (including for work in progress).

8. Change Orders.

This SOW is based on the information available and circumstances existing as of the SOW Effective Date, including the outlined Assumptions. If there are changes to such information and/or circumstances or to any other provision in this SOW, the parties will negotiate in good faith a reasonable and equitable adjustment to the fees, schedules, and/or other relevant terms and conditions set forth herein and mutually agree to a Change Order.

9. SOW Disagreement Resolution Procedures.

A. Resolution Levels.

i) **Level One Resolution.** In the event that the parties have a dispute regarding the scope of work or either party's fulfillment of its obligations of or compliance with this SOW (including any exhibits, appendices, annexes or other attachments made a part hereof to which Toptal and Client have both agreed in writing), Client's Contact and Toptal's Delivery Manager (or their respective delegates) will promptly, within no less than five (5) business days of either party's written request (email sufficient), engage in good faith discussions to attempt to resolve the dispute.

ii) **Level Two Resolution.** If such representatives are unable to resolve such dispute or have not made meaningful progress towards a resolution within such five (5) business days after such written notice, either party may request escalation and the managers of each such representative (or the delegates) will, together with the initial representatives, have an additional five (5) business days to attempt to resolve the dispute in good faith.

iii) **Level Three Resolution.** If the parties are unable to resolve the dispute within a reasonable time or have not made meaningful progress towards a resolution at the conclusion of the Level Two Resolution, either party may request a final escalation, in which case executives from each party will use good faith efforts to promptly resolve the dispute.

B. Failure to Achieve Mutually Agreed Resolution. If such final resolution efforts are unsuccessful after Level Three Resolution, subject to the provisions of Agreement and this SOW, either party may pursue those remedies available to it under the Agreement.

C. Limitations of Resolution Process. Each party will engage in good faith efforts at resolution, as described in this SOW. However, nothing in this SOW will obligate either party to enter into any particular agreement or to accept terms of resolution to which it does not accept in its sole discretion. For clarification, notwithstanding the foregoing dispute process, nothing

in this SOW will prohibit either party from seeking any remedy, at law or equity, with respect to a breach of the Agreement by the other party.

10. Excusable Delays. Toptal will not be responsible or liable for and will be excused from any delay or failure to perform under this SOW to the extent resulting directly or indirectly from causes beyond Toptal's reasonable control, including without limitation from acts of God, civil or military authority, war, riots, civil disturbances, accidents, fire, earthquakes, flood, strikes, lockouts, labor disturbances, court or governmental order, or pandemics. Without limiting the generality of the foregoing, Client acknowledges and agrees that Toptal will be excused from delays, interruption or failure in performing under this SOW to the extent resulting from Client's delay, interruption or failure in (a) providing necessary information, equipment or facility access, (b) completing its required tasks, or (c) otherwise performing its obligations under this SOW, as well as Client's buyout during the term of this SOW of any Talent providing some or all of the Scope of Work hereunder and any Assumptions in this SOW which are untrue, incorrect or incomplete.

[Remainder of page intentionally left blank. Signature page immediately follows.]

IN WITNESS WHEREOF, Toptal and Client have executed this SOW as of the SOW Effective Date.

Toptal, LLC

Client

By:

By:

Name: Carlos Aguirre

Title: CFO

Name:

Title:

EXHIBIT A

ARC Genomics Platform - Implementation Specifications

This Exhibit A is incorporated by reference into the Statement of Work to which it is attached (the "**SOW**") and forms part of the Scope of Work performed thereunder. Capitalized terms used herein and not otherwise defined have the meanings given to them in the SOW or the Agreement referenced therein. In the event of any conflict between this Exhibit and the terms of the SOW, the terms of the SOW will control unless expressly stated otherwise.

1. OVERVIEW

Under this Exhibit A, Toptal will build part of the ARC application and deliver the quality assurance (QA) test automation for the ARC program across four sequential Milestones: P1 Foundation, P2a Services and Data, P2b Application Surface, and P3 Launch. ARC is one application within a larger Client platform. Client owns the platform layer (including Terraform and infrastructure) and the data tier (including the ingestion pipelines and the Snowflake data model); Toptal owns ARC application development, comprising the micro-frontend shell, the ARC micro-frontend, the FastAPI services, and the nomenclature service. Toptal will also write and execute the automated test scripts as defined in this Exhibit. Each Milestone is an acceptance gate with its own Work Output, Acceptance Criteria, and Sign-Off, and user acceptance testing (UAT) runs after each Milestone. Story references (e.g., HS-4762) identify the corresponding items in Client's estimation sheet.

Toptal will perform the Work in this phase using methodology and terminology aligned with the Scaled Agile Framework (SAFe). Unless otherwise expressly defined in this Exhibit A or the SOW, agile and product-development terms used herein (e.g., "epic," "feature," "user story," "MVP," "backlog") have the meanings given by SAFe (scaledagileframework.com). For terms not addressed by SAFe, industry-standard definitions apply.

2. ACTIVITIES

Toptal will perform the following Work organized into the following **phases**, aligned to the Milestones in Section 4. Bullets describe the nature of the Work performed in each phase; specific Work Output and Acceptance Criteria are defined in Section 4.

2.1. P1 Foundation

- Build the micro-frontend shell, including global navigation, application routing, and remote mounting for future genomics applications, adopting Client's existing design system and making it available to remotes (HS-4762).
- Integrate the Azure AD / MSAL authentication flow into the shell and propagate it to remotes (HS-4763).
- Build and execute an end-to-end test harness covering shell composition, navigation, and authentication flows (HS-4765).
- Build automated pipeline quality gates and deployment smoke checks (HS-4770).
- Perform negative and abuse-case authentication testing and security scanning (HS-4938).
- Validate referential integrity and query patterns of the Client-built Snowflake data model (HS-4782).
- Validate Client-built data governance controls and audit classification coverage (HS-

4941).

2.2. P2a Services and Data

- Implement FastAPI faceted search, filter, and detail endpoints over the Snowflake ARC tables (HS-4777).
- Implement asynchronous processing for long-running queries, including a task queue (Celery/RQ) and health checks (HS-4778).
- Perform backend integration testing against the Snowflake development environment and load testing against the sub-second search SLA (HS-4779).
- Perform a performance engineering spike covering benchmarking, query observability, p95 latency definitions, and a tuning backlog (HS-4942).
- Build the OneGWAS direct-write integration that writes completed OneGWAS results (variant and PHG/GATSBY) to Snowflake ARC with retryable error handling (HS-4803), and execute end-to-end and failure/retry tests (HS-4804).
- Validate Client-built data ingestion and migration, covering GWAS Atlas ETL import coverage and schema-change handling (HS-4797), MTA Store migration data quality including domain-user comparison, a data-quality issue log, and sign-off support (HS-4785), and PHG/GATSBY haplotype ingestion linkage and completeness (HS-4807).
- Validate PubMed/literature extraction accuracy on key fields with an end-to-end run (HS-4801), and validate trait taxonomy coverage and approvals (HS-4791).

2.3. P2b Application Surface

- Build the ARC micro-frontend faceted search (crop, trait, marker, significance, source, and haplotype type) with a paginated, sortable results table (HS-4772).
- Build the result detail view (markers, trials, haplotypes, and study) and PHG/GATSBY interval-based visualization (HS-4773).
- Build the haplotype-type filter, result indicator, interval-based detail, and variant-versus-pangenome comparison (HS-4809), and the supporting haplotype search/filter endpoints (HS-4810).
- Implement backend endpoints and data access for haplotype association results (HS-4813).
- Implement the nomenclature service, including trait resolution, synonym lookup, unmapped-term handling, and ARC backend integration (HS-4793).
- Build and execute frontend unit, integration, accessibility, and performance tests (HS-4775), backend haplotype integration tests (HS-4815), nomenclature service tests (HS-4794), and haplotype search and visualization tests (HS-4811).

2.4. P3 Launch

- Build and execute integration, performance, security, and cross-browser (Chrome and Edge) test suites covering all data flows and UI/API critical paths in staging at production-like volume (HS-4825).
- Provide test data, fixtures, and environment support for integration testing (HS-4826).

- Execute UAT cycles with Client's scientist groups, and triage and track defects (HS-4828).
- Perform a pre-launch hardening and stabilization iteration after integration testing and UAT to close critical and high defects before cutover (HS-4943).
- Execute production smoke tests and post-deployment validation, including a 48-hour defect watch (HS-4832).
- Confirm MTA Store feature parity and zero legacy traffic before decommission (HS-4788).
- Produce training materials and user documentation (HS-4829).

3. QUALITY ASSURANCE AND TEST AUTOMATION

Toptal will deliver the QA test automation Work Output listed in Section 4. The following principles apply to all QA Work under this Exhibit A:

- **Test scope.** Testing ARC means testing against the full platform. Single sign-on (SSO) and the external integrations are within the test scope.
- **Form of Work Output.** QA Work Output consists of automated test scripts, their execution, and the resulting defect reports, rather than manual QA passes.
- **Remediation.** Where a test fails against a Client-built component, pipeline, dataset, or environment, remediation of that failure is Client's responsibility. Toptal's delivery of a test for a component does not constitute a commitment by Toptal that the component will meet its target.
- **Certification Acceptance Criteria.** Acceptance Criteria labeled "Certification" in Section 3 apply to components built by Client. For each such criterion, Toptal's Work Output is the test scripts, their execution, and a defect report measured against the stated target. Achievement of the target itself is Client's responsibility and is not a condition of Acceptance of Toptal's Work Output.
- **Re-testing.** Following Client's remediation of a failed Certification target, Toptal will re-execute the applicable test scripts in accordance with the Assumptions outlined in this contract. Additional re-runs will be addressed by Change Order.
- **Test script maintenance.** Maintenance of test scripts means in-flight upkeep of Toptal's test scripts during development. It does not include a post-launch maintenance phase.
- For test-suite Work Output (HS-4765, 4775, 4779, 4794, 4804, 4811, 4815, 4825), Acceptance requires delivery of the scripts, execution in the target environment, and a results report against the stated targets. Achievement of the targets is assessed under the corresponding build story, not the test deliverable.

4. WORK OUTPUT

Certification Acceptance Criteria. Acceptance Criteria labeled **Certification** measures a component built by Client, as described in Section 1.1. For those criteria, Acceptance of Toptal's Work Output requires delivery and execution of the test scripts and issuance of the defect report; the stated target is Client's to achieve and remediate. All other Acceptance Criteria measure Toptal-built Work.

Latency Acceptance Criteria. Latency and render-time Acceptance Criteria measure Toptal-built components running on Client-owned platform components. These include Snowflake (warehouse sizing and configuration, schema, views, and data volume), Azure AD, CloudFront, Client's upstream systems (including OneGWAS), and Client's network and environments. Telemetry, including the query observability delivered under HS-4942, may show that a missed target is attributable to a Client-owned platform

component or Client-provided third-party item (including Cogen and Tagasys), or to data volumes, usage patterns, or environmental conditions outside Toptal's reasonable control, rather than to Toptal-built code or Toptal-authored queries. In that case, the criterion will be treated as met for Toptal's Work Output, and the finding will be recorded in the defect report for Client remediation.

Milestone	Story	Work Output	Acceptance Criteria	Type
Milestone 1: P1 Foundation <i>Est. weeks 1–6</i>	HS-4762	Micro-frontend shell with global navigation, routing, built with Client's design system and sharing it with remotes	Shell load P50 < 1.5s, P90 < 2.5s, P95 < 3s; remote mount P95 < 1s	Toptal · Latency
	HS-4763	Azure AD / MSAL authentication integration, propagated to remotes	Authentication success 100%; token acquisition P95 < 800ms; silent token refresh works	Toptal · Latency
	HS-4765	Shell and authentication-flow E2E test harness	E2E coverage of composition, navigation, and authentication; 100% critical-path pass	Toptal
	HS-4770	Pipeline quality-gate and deployment smoke-check automation	Executed against target: quality gates block on fail; deploy smoke 100% pass before promote	Certification
	HS-4938	Authentication negative-test suite and security scan results	Negative tests pass; scan clean; findings in Toptal-built components remediated, findings in Client-built components reported to Client	Toptal
	HS-4782	Data model validation scripts, results, and defect report	Executed against target: referential integrity 100%; representative query patterns validated	Certification
	HS-4941	Data governance validation scripts, audit results, and gap log	Executed against target: controls audited; classification coverage validated; gaps logged	Certification
Milestone 2: P2a Services and Data <i>Est. weeks 7–16</i>	HS-4777	Faceted search and detail endpoints	Search P50 < 300ms, P90 < 700ms, P95 < 1s; detail P95 < 500ms	Toptal · Latency
	HS-4778	Asynchronous long-running query processing with health checks	Long queries offloaded; status poll P95 < 200ms; queue drains; health checks green	Toptal · Latency
	HS-4779	Backend integration and load test suites and results	Search P95 < 1s.	Toptal · Latency

Milestone	Story	Work Output	Acceptance Criteria	Type
	HS-4942	Performance spike output: benchmarks, query observability, p95 definitions, and tuning backlog	p95 targets defined; query instrumentation implemented; tuning backlog triaged	Toptal · Latency
	HS-4803	OneGWAS direct-write integration with retryable error handling	100% of completed runs written; retryable errors recovered; write P95 < 60s	Toptal · Latency
	HS-4804	OneGWAS E2E and failure/retry test suite and results	Completion to ARC searchable P95 < 5 min; failure/retry validated	Toptal · Latency
	HS-4797	GWAS Atlas ETL validation scripts, results, and defect report	Executed against target: ≥90% import coverage; schema-change handling verified	Certification
	HS-4785	MTA Store migration validation, data-quality issue log, and sign-off support	Executed against target: domain-user comparison sign-off; data-quality issue log triaged/closed	Certification
	HS-4807	PHG/GATSBY ingestion validation scripts, results, and defect report	Executed against target: linkage and completeness validated (100% sampled keys match)	Certification
	HS-4801	PubMed/literature extraction validation scripts, results, and defect report	Executed against target: ≥85% precision on key fields; E2E run green	Certification
	HS-4791	Trait taxonomy validation scripts, results, and defect report	Executed against target: coverage ≥80%; approvals captured	Certification
Milestone 3: P2b Application Surface <i>Est. weeks 17–21</i>	HS-4772	Faceted search and results table	Results render P50 < 400ms, P90 < 800ms, P95 < 1s (excl. network)	Toptal · Latency
	HS-4773	Result detail view and interval-based visualization	Detail view P90 < 1s; interval visualization render P95 < 1.5s	Toptal · Latency
	HS-4775	Frontend unit, integration, accessibility, and performance test suites	Unit coverage ≥80%; key-flow integration green; search < 1s	Toptal · Latency
	HS-4809	Haplotype filter, indicator, interval visualization, and variant-versus-pangenome comparison	Filtered results P90 < 1s; interval visualization P95 < 1.5s; comparison works	Toptal · Latency

Milestone	Story	Work Output	Acceptance Criteria	Type
	HS-4810	Haplotype search/filter endpoints	Filter endpoint P95 < 1s	Toptal · Latency
	HS-4813	Haplotype API and data access	Endpoints P95 < 1s; correct results against fixtures	Toptal · Latency
	HS-4815	Haplotype endpoint integration tests and results	Integration tests green (100% of critical paths)	Toptal
	HS-4793	Nomenclature service	Resolution correct for all mapped traits; lookup P95 < 200ms	Toptal · Latency
	HS-4794	Nomenclature test suite and results	100% mapped-trait resolution correct; lookup P95 < 200ms	Toptal · Latency
	HS-4811	Haplotype search and visualization tests and results	Filters, comparison, and rendering validated across haplotype types	Toptal
Milestone 4: P3 Launch <i>Est. weeks 22–26</i>	HS-4825	Integration, performance, security, and cross-browser test suites and results	All flows executed at production-like volume; search P95 < 1s; authentication/security pass; Chrome and Edge validated; failures in Client-built components recorded in the defect report	Toptal · Latency
	HS-4826	Test data and fixtures	Test data and fixtures cover all integration scenarios; environments supported	Toptal
	HS-4828	UAT execution records and defect log	UAT cycles run; defects triaged and tracked; no open critical defects in Toptal-built Work at exit	Toptal
	HS-4943	Hardening iteration results, including full regression run	All critical and high defects in Toptal-built Work closed; full regression green before cutover	Toptal
	HS-4832	Production smoke test results and 48-hour defect watch report	Production smoke 100% pass; 48-hour defect watch clean	Toptal
	HS-4788	MTA Store parity and usage-zero confirmation report	Executed against target: parity confirmed; legacy traffic = 0 over watch window before decommission	Certification
	HS-4829	Training materials and user documentation	Training materials and user documentation delivered and reviewed	Toptal

5. ENGAGEMENT GOVERNANCE

5.1. Engagement Cadence

This phase of work will maintain the following cadence:

- **Kickoff Call (Day 1):** full-team alignment session to review phase governance, confirm cadenced meeting times, and discuss outstanding access provisioning and documentation (if applicable).
- **Biweekly Sprint Demo:** review work completed in past sprint, review Milestone status, open risks/issues, and decisions needed from the Client.
- **Daily Standups:** Toptal team will join Client's daily standups to track progress and collaborate with the Client-side team members.
- **Ad-Hoc Working Sessions:** scheduled as needed throughout the engagement to align on open questions that pertain to the delivery of this scope of work.
- **Milestone Acceptance Review:** held at the end of each milestone. Covers milestone acceptance, Work Output review, and forward-looking risk assessment for the next milestone.

5.2. Weekly Report

The Delivery Manager will send a Weekly Status Report to the designated Client Contact after the Weekly Status Meeting. The Client Contact will be asked to send their confirmation of acceptance and/or any additional feedback in response to each report. The report will cover:

- Milestone billing snapshot
- Milestone progress
- Work Output status for the active Milestone
- Open risks & issues
- Open blockers & decisions needed

6. CLIENT RESPONSIBILITIES

The Client shall provide the following:

- All Client code dependencies and Client-owned stories must be completed before the related Toptal-owned Work can begin.
- Make the following Client personnel available for the Toptal stories that depend on them: domain users; domain experts for trait taxonomy; scientist UAT groups; and Client's identity provider (IdP) team for Azure AD / MSAL authentication.
- Maintain the development, staging, and production environments and resolve environment blockers throughout all Milestones, including P2a and P2b.
- Remediate defects identified by Toptal's testing in Client-built components, pipelines, datasets, and environments.
- Provide access to the design system (code library, tokens, design files, guidelines and a named contact), and UI/UX designs for the Milestone 3 screens.
- Provide Milestone Sign-Off through Client's designated approvers,
- Coding standards and delivery documentation standards (merge request process, code review gates, unit and integration test requirements) are required on or before the start date.

- Timely written response to decision requests flagged in weekly status reports.
- Timely written response to provide confirmation and/or feedback upon receiving each weekly status report.

7. ASSUMPTIONS & DEPENDENCIES

- The Milestones run sequentially as acceptance gates. P2a Services and Data begins upon completion of HS-4781; P2b Application Surface begins upon acceptance of P2a; and P3 Launch begins upon acceptance of P2b.
- The Cogen API is assumed to be available and stable at approximately 95%, as assessed by Syngenta. Test failures and re-test cycles arising from Cogen availability below that level will be addressed by Change Order.
- The estimated schedule, measured from the Start Date, is P1 weeks 1–6, P2a weeks 7–16, P2b weeks 17–21, and P3 weeks 22–26.
- The schedule and cost were created considering that no breaks or stoppages due to holidays will occur to impact delivery.
- UAT runs after each Milestone and is included within the fixed price. UAT is limited to two (2) runs per Milestone, covering all UAT and not only Toptal-built Work. UAT runs beyond two will be addressed by Change Order.
- Changes outside Toptal's control that require additional UAT or test cycles which result in rework of Toptal's test scripts will be addressed by Change Order. Such changes include upstream Snowflake schema changes and API taxonomy shifts.
- Client owns the platform layer (including Terraform and infrastructure) and the data tier (including the ingestion pipelines and the Snowflake data model). The platform shell hosts CloudFront and Active Directory.
- ARC reads Snowflake tables populated by upstream systems (MTA, 1GWAS, GATS-B, and GWAS Atlas), all of which are within Client's control.
- The external API integrations within scope are Cogen and Tagasys.
- Toptal will use Client's existing design system for all user-facing screens. Estimates include integrating it and building up to [N] ARC-specific components (e.g., interval visualization, facet panel), and exclude creating, extending or maintaining the design system. Further components or changes are addressed by Change Order.
- By the Start Date, the Client will provide access to the design system (code library, tokens, design files, guidelines and a named contact), and UI/UX designs for the Milestone 3 screens. The Client will give advance notice of design-system changes; resulting rework is addressed by Change Order.
- Toptal will write and execute the automated test scripts listed as Work Output in Section 4, covering Toptal-built Work and the Client-built items identified there as Certification. Testing of other Client-built components, pipelines or environments is not included and may be added by Change Order.
- Toptal's Milestone schedule and fixed fees assume that Client completes the Client-built work, personnel availability, environment support, reviews, and approvals described in Section 5 before the dependent Toptal Work is scheduled to begin.

8. THIRD PARTY ITEMS

The following Third Party Items are used during the ARC engagement, and the party providing or making each item available is indicated below. Client's post-delivery license responsibilities are governed by the Third Party Items section of the SOW.

- **Snowflake** — Data platform hosting the ARC tables that Toptal's services read and write and against which Toptal's tests execute. Provided by Client during the engagement.
- **Azure AD / MSAL** — Identity provider and authentication library used for shell and remote authentication. Provided by Client during the engagement.
- **Azure Key Vault** — Secrets management used by the CI/CD pipelines. Provided by Client during the engagement.
- **CloudFront** — Content delivery hosted by the Client platform shell. Provided by Client during the engagement.
- **Cogen and Tagasys APIs** — External API integrations within the test scope. Provided by Client during the engagement. [confirm]
- **Playwright** — Open-source framework for end-to-end testing of the ARC React/Next.js front end, and for UI and API test automation of the Python services where applicable. Provided by Client during the engagement.
- **Jest** — Open-source JavaScript test framework for component and integration testing of the ARC front end, including code coverage reporting. Provided by Client during the engagement.
- **React Testing Library** — Open-source library used with Jest for component and integration testing of the ARC front end. Provided by Client during the engagement.
- **Pytest** — Open-source primary test framework for the Python (FastAPI) services, including the nomenclature service. Provided by Client during the engagement.

9. OUT OF SCOPE

The following are excluded from Toptal's Work under this Exhibit A. Except where noted, each item is performed by Client. Toptal's QA test automation for these items, where listed in Section 3, remains in scope.

- Platform infrastructure and DevOps, including environment and IaC strategy, Terraform modules, CI/CD pipelines, monitoring and alerting, and shell telemetry and observability (HS-4766, HS-4767, HS-4768, HS-4769, HS-4764).
- Security design, threat modeling, platform hardening, and RBAC/secrets implementation (HS-4936, HS-4937).
- Architecture and design stories, including the shell architecture and onboarding contract, ARC frontend component architecture, API design and OpenAPI contract, OneGWAS output contract, nomenclature service interface design, haplotype search/visualization design, and backend haplotype support design (HS-4761, HS-4771, HS-4776, HS-4802, HS-4792, HS-4808, HS-4812).
- Snowflake data model design, schema DDL, schema extensions for haplotypes, and data governance design and controls (HS-4780, HS-4781, HS-4814, HS-4939, HS-4940).
- Data ingestion and migration build, including MTA Store migration design and scripts, GWAS Atlas ETL, PubMed/literature ingestion, PHG/GATSBY haplotype ingestion, and trait taxonomy definition

(HS-4783, HS-4784, HS-4795, HS-4796, HS-4798, HS-4799, HS-4805, HS-4806, HS-4789, HS-4790).

- Integration test plan (HS-4824) and UAT coordination and sign-off (HS-4827).
- Go-live planning and operational runbook handover, production deployment, MTA Store decommission planning, and legacy infrastructure teardown and redirects (HS-4830, HS-4831, HS-4786, HS-4787).
- Hypercare and post-go-live support beyond the 48-hour post-deployment defect watch (HS-4944), and any post-launch maintenance phase.
- Remediation of defects in Client-built components, pipelines, datasets, or environments.
- Achievement of numeric targets for Client-built work, including PubMed extraction precision, GWAS Atlas import coverage, trait taxonomy coverage, Snowflake referential integrity, CI/CD quality gates, and decommission parity.
- Creation, extensions, or maintenance of the Client-provided design system.